

$$F = \text{grad } f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (F_1, F_2, F_3)$$

$$1) \frac{\partial f}{\partial x} = F_1 = ye^x \sin z \rightarrow \int \frac{\partial f}{\partial x} dx = \int ye^x \sin z dx$$

$$\rightarrow f(x, y, z) = ye^x \sin z + \alpha(y, z)$$

$$2) \frac{\partial f}{\partial y} = F_2 = 1 + \cancel{e^x \sin z} = \frac{\partial}{\partial y} (ye^x \sin z + \alpha(y, z))$$

$$= \cancel{e^x \sin z} + \frac{\partial \alpha}{\partial y}, \quad \frac{\partial \alpha}{\partial y} = 1 \rightarrow \alpha(y, z) = y + \beta(z)$$

$$3) \frac{\partial f}{\partial z} = F_3 = \cancel{ye^x \sin z} = \frac{\partial}{\partial z} (ye^x \sin z + \underbrace{y + \beta(z)}_{\alpha(y, z)})$$

$$f(x, y, z) = ye^x \sin z + y + \beta(z)$$

$$= \cancel{ye^x \sin z} + 0 + \beta'(z), \quad \beta'(z) = z \rightarrow \beta(z) = \frac{1}{2} z^2 + C, \quad C \in \mathbb{R}.$$

$$f(x, y, z) = y e^x \sin z + y + \frac{1}{2} z^2 + c$$

Note:

$$G_1 \text{ is s.t. } \frac{\partial G_1}{\partial x} = F_1$$

$$\int \frac{\partial f}{\partial x} dx = \int F_1 dx, \quad f(x, y) = G_1 + \alpha(y)$$

$$\int \frac{\partial f}{\partial y} dy = \int F_2 dy, \quad f(x, y) = G_2 + \beta(x)$$

2.3.4 Proof of Theorem 1 (for $n=2$)

24/09/2026

a) Necessary condition.

If F derives from a potential in Ω and $F \in C^1(\Omega, \mathbb{R}^2)$

$$\Rightarrow f \in C^2(\Omega) \text{ s.t. } \text{grad } f = F$$

$$\text{curl}(\text{grad } f) = \text{curl}(F) = 0 \quad \forall (x,y) \in \Omega.$$

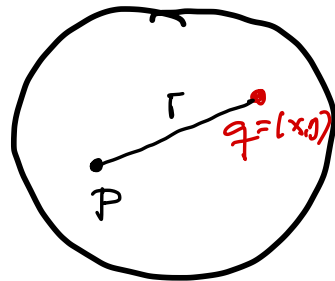
b) Sufficient condition (only convex domains, limited version).

- Assume that $P = (0,0)$ is in Ω .

- the segment Γ joining a couple of points $P, q \in \Omega$ is $\Gamma \subset \Omega$ (Ω is convex)

- $\text{curl } F = 0$

Find $\varphi : \Omega \rightarrow \mathbb{R}$, s.t. $\text{grad } \varphi = F$



Ansatz:

$$\varphi(x, y) = \int_{\Gamma(x, y)} \mathbf{F} \cdot d\mathbf{l} = \int_0^1 \mathbf{F}(\gamma(t)) \cdot \gamma'(t) dt$$

$$\Gamma(x, y)$$

$$\gamma: [0, 1] \rightarrow \mathbb{R}^2$$

$$t \mapsto \gamma(t) = (tx, ty)$$

$$\gamma'(t) = (x, y)$$

$$\varphi(x, y) = \int_0^1 (\mathbf{F}_1(\gamma(t)), \mathbf{F}_2(\gamma(t))) \cdot (x, y) dt$$

$$= \int_0^1 x \mathbf{F}_1(\gamma(t)) + y \mathbf{F}_2(\gamma(t)) dt$$

$$\mathbf{F} = \text{grad } \varphi = \left(\frac{\partial \varphi}{\partial x}, \frac{\partial \varphi}{\partial y} \right)$$

$$\frac{\partial \varphi}{\partial x} = F_1 = \int_0^1 F_1(tx, ty) + xt \frac{\partial F_1}{\partial x}(tx, ty) + yt \frac{\partial F_2}{\partial x}(tx, ty)$$

$$\frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$$

$$= \int_0^1 F_1(tx, ty) + xt \frac{\partial F_1}{\partial x}(tx, ty) + yt \frac{\partial F_1}{\partial y}(tx, ty) dt$$

$$= \int_0^1 F_1(tx, ty) + t \underbrace{\text{grad } F_1(tx, ty) \cdot (x, y)}_{\gamma'(t)} dt$$

$$\frac{d}{dt} (t F_1(\gamma(t)))$$

$$= \frac{d}{dt} (t F_1(tx, ty))$$

$$= \int_0^1 \frac{d}{dt} (t F_1(\gamma(t))) dt = t F_1(\gamma(t)) \Big|_{t=0}^{t=1}$$

$$= 1 \cdot F_1(x(1)) - 0 \cdot \cancel{F_1(x(0))} = F_1(x, y)$$

$$\frac{\partial \varphi}{\partial x} = F_1(x, y).$$

I do the same for $\frac{\partial \varphi}{\partial y} = F_2(x, y)$

$$\text{grad } \varphi = \left(\frac{\partial \varphi}{\partial x}, \frac{\partial \varphi}{\partial y} \right) = (F_1(x, y), F_2(x, y)) = F(x, y).$$

□

with $\varphi = \int_{\gamma} F \cdot dl$

2.3.2 Examples (continuation)

Example 2: let be a vector field

$$F: \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow \mathbb{R}^2$$

$$(x,y) \mapsto F(x,y) = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right)$$

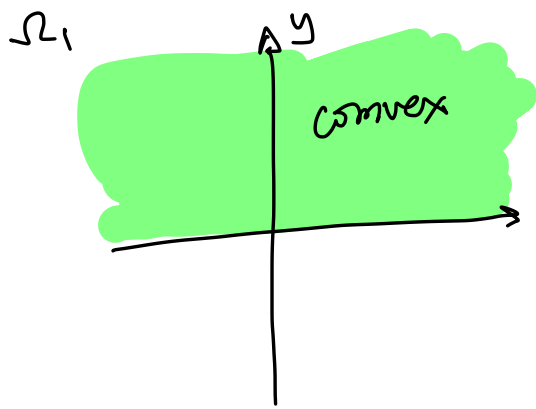
and we consider the domains:

$$\Omega_1 = \{ (x,y) \in \mathbb{R}^2 : y > 0 \}, \quad \Omega_2 = \{ (x,y) \in \mathbb{R}^2 : y < 0 \}$$

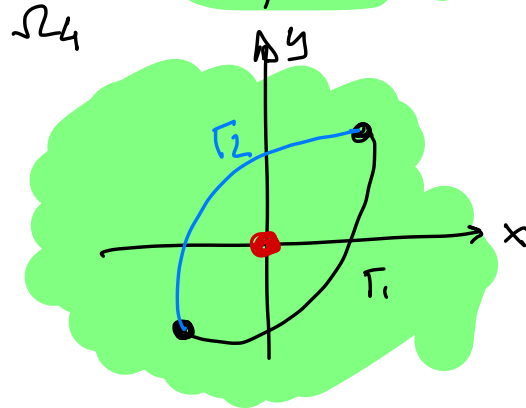
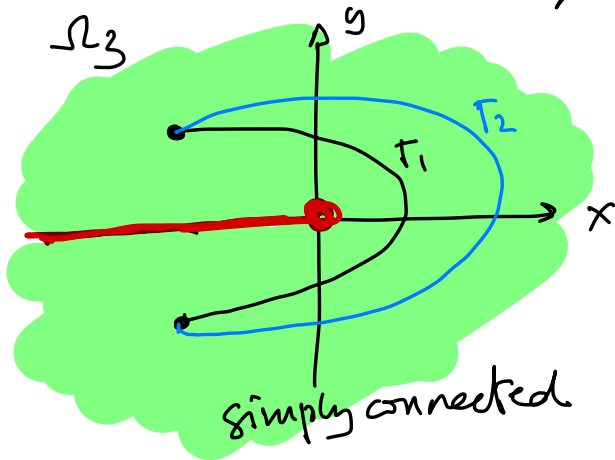
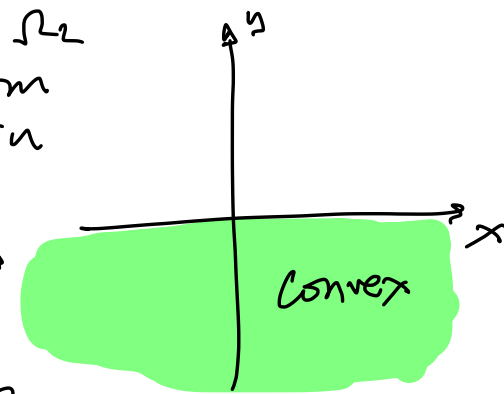
$$\Omega_3 = \mathbb{R}^2 \setminus \{ (x,y) \in \mathbb{R}^2 : x \leq 0 \text{ and } y = 0 \}$$

both conditions
should be met at
the same time

$$\Omega_4 = \mathbb{R}^2 \setminus \{(0,0)\} = \text{dom}(F)$$



F derives from
a potential in



non-simply connected.

$\text{curl } F = 0 ? \rightarrow \text{YES.}$